

Beyond the sector: audit the time support

Optional extension to the feature and observation-support audits

The rose summarizes a signal over an interval. This exercise asks what the summary preserves and what it discards. Consider the 40-second, 100-Hz synthetic three-axis example described below and on the reference page. Record predictions before consulting the reference values. Its acceleration units are arbitrary (au), not calibrated g. It is **separate from the 800-position schematic bins in Figs. 1–3**.

1. Same statistic, different temporal account

Compare the complete synthetic signal with a copy shifted by 20 seconds using a circular sample permutation. They contain exactly the same vectors in a different order. Before consulting the reference page, predict whether whole-record RMS, one-second summaries, and the location of the strongest burst will change.

Decision to record. Can the same rose sector support both reports' claim about *when* movement occurred? Identify the temporal evidence you would request. Do not infer stationarity from a single whole-window statistic.

2. Mean square is not automatically variance

Consider the three axis means and the sum of their centered population variances for a quiet one-second interval and an active interval. Ask whether a nonzero local mean contributes to mean square. State what preprocessing would be needed before calling the feature “variance”.

3. Missing acquisition or excluded observation?

The fixture has no acquired readings from 15 to 18 seconds. Elsewhere, a deliberately configured sensor range clips some readings; the toy quality rule flags these *recorded* values as unsuitable. Keep the two reasons separate. The clipping rule is an instructional choice, not a validated detector of wearable artifacts.

Compare two further constructed masks that each omit 20% of samples: one drops the largest squared magnitudes, the other the smallest. Does equal coverage establish equal retained information? What should accompany a reported coverage fraction?

4. Average magnitude or time accumulation?

Compare one complete 40-second record with the same record repeated to 80 seconds. Which feature should remain unchanged: RMS, mean square, or the time integral of squared acceleration? State the units and explain why the integral is not mechanical or metabolic energy.

Return to the report. Add the sampling rate, analysis window, display aggregation, units, validity rule, and permitted interpretation to your audit. For irregular sampling, request an explicit support-weight and gap policy. Never let a timestamp gap silently become a long interval assigned to its preceding sample.

Instructor reference and implementation

Release after predictions and rationales have been recorded

Synthetic signal construction

The piecewise amplitude and instantaneous frequency create changing local moments. The fixture is deterministic: it demonstrates nonconstant local summaries, not a statistical stationarity test. A 0.30-au offset on the first axis keeps the mean-square/variance distinction visible. Latent values are known only because the fixture is constructed; they do not represent recoverable missing measurements.

Reference distinctions

For supplied positive time-support weights Δt_n and validity indicators v_n , let the observation support be $T_{\text{obs}} = \sum_{v_n=1} \Delta t_n$. Then

$$E_{\text{obs}} = \sum_{v_n=1} \|\mathbf{u}[n]\|_2^2 \Delta t_n, \quad M_{2,\text{obs}} = E_{\text{obs}}/T_{\text{obs}}, \quad R_{\text{obs}} = \sqrt{M_{2,\text{obs}}}.$$

These are finite sampled-support quantities. E_{obs} has units $\text{au}^2 \text{ s}$. It approximates a continuous-time squared-acceleration integral only under a stated sampling/reconstruction convention. It is not in joules. $M_{2,\text{obs}}$ and R_{obs} remain undefined when $T_{\text{obs}} = 0$.

With population normalization on the *same support*, vector-magnitude mean square equals the sum of centered axis variances plus the squared norm of the axis-mean vector. It does not equal the variance of vector magnitude. Use explicit support weights and do not interpolate across missing support.

Contrast	Reference decision
20-second circular shift	Whole-record mean square, RMS, and squared integral are unchanged; one-second summaries move in time. A whole-record scalar cannot locate the burst.
40 versus 80 seconds	Repetition preserves RMS and mean square but doubles squared integral: approximately 30.1203 to 60.2405 $\text{au}^2 \text{ s}$.
Nonzero mean	Full-record mean square is approximately 0.7530 au^2 ; the sum of centered axis variances is approximately 0.6537 au^2 . The remainder is squared mean.
Same 80% coverage	Removing the largest versus smallest magnitudes yields different observed means. Coverage alone is not a missingness model.
Unequal support weights	For axis values 1 and 3 with weights 0.1 and 0.9 seconds, weighted mean square is 8.2, not the sample-count average 5.0.

Classroom use

Use the first page for individual predictions or paired discussion. Release the reference page after learners have identified the window, units, validity rule, and evidence needed for their interpretation. Keep acquisition gaps separate from quality exclusions in the decision record. A paper-based activity can use the comparisons and reference values above without executing code.